Answers and talk notes

Determine Surface Area of Prisms

What your student is learning

Tonight we find the surface area of a triangular prism — a tent shape — by adding its two triangle ends and its three rectangles.

Answers

Ask your student to explain before you look.

1. **B.** 2 triangles and 3 rectangles. 2 triangles and 3 rectangles make the five faces.
2. **A.** 12 square units. 12 square units: ½ × 6 × 4 = 12.
3. **D.** 40. 40: the rectangle is 4 by 10, and 4 × 10 = 40.
4. **C.** 132 square units. 132 square units: 6 + 6 + 30 + 40 + 50 = 132.

Talk about it

Which faces use the length of the tent as one of their sides?

Home mission

Find a tent-shaped box, like a snack package or a toy tent. Count the faces out loud: triangles first, then rectangles. Which rectangle is the biggest?

When they are stuck, ask

- How many faces does this solid have? Name them one by one.
- Which side do all three rectangles share?
- Are we covering the outside or filling the inside?

Try not to say

- Multiply length by width by height for surface area.
- One triangle is enough, just remember the other is the same.
- Skip the bottom rectangle, nobody sees it.